Luke Abraham

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EDUCATION

Texas A&M University, College Station, Texas — GPA: 3.95

B.S. Computer Science, minor in Mathematics — *May 2028*

- **Relevant Coursework:** Design and Analysis of Algorithms**,** Computer Architecture, Statistics, Machine Learning Specialization

EXPERIENCE

Amazon Web Services — **September 2026 - December 2026**

Software Development Engineer Intern

- Developing on AWS low-latency microservices and agentic systems, reaching **4 million+ customers** daily

SpaceX — **May 2026- August 2026**

Software Engineer Intern

- Developed an optimization algorithm to improve Starship low-level **3-D** rendering performance, enabling **5,000+ engineers** to visualize **millions of Starship components and assemblies** with over **99% reductions in polygon counts**
- Designed custom scalable CI/CD systems with **Kubernetes, Docker,** and **Kafka** (C#, C++, Bash, ASP.NET, React.js, Linux, Bazel)
- Working on reinforcement learning with SpaceXAI for to scale Grok's engineering capabilities

Texas A&M University – Reddy Lab — **February 2026 - May 2026**

Data Science Researcher

- Analyzed biomarker data of **100,000+ datapoints**, statistically modeling the progression of post-traumatic epilepsy
- Integrated biological insights with data-driven approaches to track disease development, improving understanding of post-traumatic epilepsy and informing potential interventions for over **69 million affected individuals**

Exterro — **May 2024 - August 2024**

Machine Learning Engineer Intern

- Constructed an NLP system that classifies **50,000+ data points** for lawyers and detectives to locate critical digital evidence, achieving **97-99% accuracy** with **less than 1% manual labeling**
- Trained CNNs and LSTM networks using Python (NumPy, Pandas, scikit-learn, spaCy, TensorFlow)

PROJECTS/COMPETITIONS

IMC Prosperity 4 - Top 6% Globally — **February 2026**

- Competed solo, developing algorithmic strategies and optimizing portfolio performance across simulated markets

Hudson River Trading MetaPuzzle Competition (1st place) — **February 2026**

- Solved **8** quantitative and pattern recognition puzzles in **45 minutes**, integrating sub-puzzle solutions into a final metapuzzle to demonstrate rapid analytical reasoning and collaborative problem-solving against **100+ competitors**

Pipeline Anomaly Detection - TidalHackMachine Learning Hackathon Winner — **February 2026**

- Built an end to end pipeline integrity ML system for Regulatory Compliance Partners, aligning **3 ILI runs** across **15 years** of data and **980+ defects** using, spatial indexing, and anomaly detection, **reducing reconciliation time by 80%**
- Completed full-stack web app **in 24-hours** against **300+ competitors** (Python, React.js)

Cloudflare LLM Chatbot — **January 2026**

- Created a full-stack AI chat application deployed on Cloudflare Workers, leveraging **Llama 3.3 70B** for intelligent, context-aware responses with low-latency edge deployment and a minimalistic dark-themed UI with real-time typing indicators
- Engineered persistent conversation memory using SQLite-based Durable Objects to maintain context across **20+ message exchanges** per session, **reducing response latency to under 200ms** for cached edge requests

Tidal Machine Learning Competition (1st place) — **October 2024**

- Demonstrated expertise across **100+ competitors** in a comprehensive examination of machine learning theory, neural network architecture, and optimization algorithms

The Universal Guide to Object Oriented Programming in Java — **March 2024**

- Authored and self-published an instructional e-book on Amazon featuring comprehensive coding examples for object-oriented programming principles
- Reached **600+ learners** across academic and self-taught communities, maintaining a **5.0-star rating** across all reader reviews

SKILLS

Programming Languages/Frameworks: Python, Java, C++, R, JavaScript, TypeScript, React, Selenium

Tools/Platforms: Git, RESTful APIs, Linux, Microsoft Office Suite, Docker, AWS, React Native, Kubernetes